# GAME SENSE

## ESPORTS PERFORMANCE & GAMING HEALTH ANALYTICS

### DATA ANALYTICS PROJECT DOCUMENTATION

Team: Data Dynamos

Institution: TEKS Academy

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Project Type: Data Analytics Project

Dataset Source: Kaggle

Data Type: Player-Session Level Esports Records

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## 1. Project Title

Game Sense – Esports Performance & Gaming Health Analytics

## 2. Project Objective

Our main goal with this project was to take raw esports match and biometric data and turn it into insights that actually matter for players and coaches. We wanted to understand how performance indicators like APM, reaction time and win rate relate to health and lifestyle habits such as sleep, energy drink consumption, screen brightness and blue-light filter use, across different game genres. We used Python and Pandas to clean and explore the data, which helped surface patterns and relationships that could support better training and recovery decisions. To make everything easy to explore and present, we brought the results together in an interactive Power BI dashboard.

## 3. Problem Statement

- Esports data contains detailed information on sleep, playtime, reaction time, APM, energy drink use, and physiological signals like heart rate and HRV — but on its own, this raw data doesn't reveal much.
- It's not clear which of these factors actually separate winning matches from losing ones.
- It's not obvious how these relationships shift across different game genres.
- There's no easy way to tell whether performance dips are more closely tied to skill-based factors (APM, reaction time) or recovery-based factors (sleep, energy drinks, heart rate/HRV) without proper analysis.
- Without proper cleaning and analysis, it's hard to make real use of the data for decision-making — which is exactly what this project set out to fix by uncovering the patterns that matter and turning them into insights players and coaches can act on.

## 4. Dataset Information

| Dataset Attribute | Details |
|---|---|
| Dataset Name | Esports Gaming Biometrics Dataset |
| Total Records | 250,000 rows |
| Total Columns | 13 columns (16 after feature engineering) |
| Data Type | Player-session level (one row per match session) |
| Data Source | Kaggle |
| Row Meaning | Each row represents one player's biometric and performance readings for a single match session |
| Important Columns | Player ID, Age, Game Genre, Playtime Hours, Sleep Hours, Energy Drinks, Screen Brightness, Blue Light Filter, APM, Heart Rate (BPM), HRV Score, Reaction Time (ms), Match Outcome |

The dataset is made up of session-level esports records, where each row represents one player's performance and biometric readings for a single match. It includes information on gameplay performance (APM, reaction time, match outcome), physiological state (heart rate, HRV), and lifestyle habits (sleep hours, energy drinks, screen brightness, blue-light filter use) — everything needed to clean, explore, and pull meaningful performance insights from the data.

## 5. Tools and Technologies Used

| Tool / Technology | Purpose |
|---|---|
| Python | Used as the primary programming language for data cleaning, analysis, and exploratory data analysis. |
| Pandas / NumPy | Used for data manipulation, cleaning, transformation, and validation of the dataset. |
| Matplotlib / Seaborn | Used for exploratory visualization during the analysis process. |
| Power BI | Used to build an interactive dashboard for visualizing performance and health insights. |

# 6. Data Cleaning

Before we could explore the data, we cleaned and validated it using Python and Pandas. The raw dataset started out with 250,000 rows and 13 columns.

## 6.1 Missing Value Handling

We found missing values in two fields — screen_brightness and hrv_score. Rather than dropping these records, we filled the missing values using the column median so the rest of the analysis could still use the full dataset.

## 6.2 Duplicate Record Handling

We checked the dataset for exact duplicate rows and did not find any that needed to be removed.

## 6.3 Column Standardization

We standardized all column names to lowercase for consistency across the cleaning and analysis workflow.

## 6.4 Data Validation

We checked all numeric columns (APM, heart rate, HRV score, reaction time, sleep hours, playtime hours) for negative or zero values as a sanity check and did not find any invalid entries.

## 6.5 Feature Engineering

We created three grouped fields to make trends easier to compare and visualize: sleep_groups (binned sleep_hours), apm_groups (binned apm), and reaction_time_group (binned reaction_time_ms).

## 6.6 Outlier Analysis

We ran an IQR-based outlier check across all numeric columns. We chose to keep the flagged values, since the dataset naturally mixes low- and high-performance sessions, so many of the "extreme" readings are legitimate high-intensity or low-recovery observations rather than data errors. Removing them would have distorted the performance picture rather than cleaned it up.

Once the data was cleaned and validated, we moved on to exploratory data analysis and building the Power BI dashboard.

# 7. Data Analysis / Exploratory Data Analysis

With the data cleaned, we used Pandas, Matplotlib, and Seaborn to explore performance patterns, health and behavioral relationships, and genre-level differences.

## 7.1 Performance Analysis

Win rate by APM group:

| APM Group | Win Rate |
|---|---|
| <100 | 63.64% |
| 100-200 | 44.91% |
| 200-300 | 49.88% |
| 300-400 | 51.71% |
| 400+ | 55.34% |

Win rate by reaction time group:

| Reaction Time Group | Win Rate |
|---|---|
| <220 ms | 50.19% |
| 220-250 ms | 54.20% |
| 250-280 ms | 50.36% |
| 280-310 ms | 46.93% |
| 310+ ms | 44.89% |

Reaction time in the 220–250 ms range had the highest win rate, and win rate dropped off steadily for reaction times above 280 ms. The lowest APM group had a notably higher win rate than the higher APM groups, showing raw action speed alone doesn't guarantee a win.

### 7.2 Health & Behavioral Analysis

Win rate by sleep group:

| Sleep Group | Win Rate |
|---|---|
| <5 Hours | 47.95% |
| 5-7 Hours | 50.71% |
| 7-9 Hours | 51.61% |
| 9+ Hours | 52.12% |

Average heart rate and HRV score by sleep/energy drinks:

| Metric | Trend |
|---|---|
| Avg. Reaction Time by Sleep Group | Falls from ~274 ms (<5 hrs) to ~247 ms (9+ hrs) |
| Avg. HRV Score by Sleep Group | Rises from ~83.8 (<5 hrs) to ~92.1 (9+ hrs) |
| Avg. Heart Rate by Energy Drinks | Rises from ~80 bpm (0 drinks) to ~97 bpm (4 drinks) |
| Blue Light Filter Usage | About 65% of players use a blue-light filter |

More sleep is consistently linked to a higher win rate, lower reaction time, and higher HRV (a marker of better recovery). Energy drink consumption shows a strong link to heart rate but not to win rate directly.

### 7.3 Genre Analysis

Win rate and average HRV score by game genre:

| Game Genre | Win Rate | Avg. HRV Score |
|---|---|---|
| Fighting | 50.63% | 87 |
| MOBA | 49.74% | 87 |
| FPS | 46.46% | 88 |
| Battle Royale | 45.61% | 88 |

Win rate varies modestly by genre, but HRV score stays fairly consistent (87–88) regardless of genre — suggesting genre choice has little effect on physiological recovery state.

## 8. Key Insights and Findings

### 8.1 Sleep Improves Win Rate

Players sleeping 9+ hours have the highest win rate (52.12%), compared to 47.95% for players sleeping under 5 hours. More sleep is also linked to lower reaction time and higher HRV, pointing to a genuine recovery effect.

### 8.2 Reaction Time Beats Raw APM

Reaction time in the 220–250 ms range wins more often (54.20%) than slower reaction times of 310+ ms (44.89%). Meanwhile, the lowest APM group had the highest win rate (63.64%) — raw action speed alone doesn't predict winning.

### 8.3 Energy Drinks Raise Heart Rate, Not Necessarily Wins

Average heart rate climbs sharply with energy drink consumption, from about 80 bpm with none to 97 bpm with 4 drinks, but this didn't translate into a clear win-rate benefit.

### 8.4 Genre Has a Modest Effect on Win Rate, Little Effect on HRV

Fighting games had the highest win rate among tracked genres (50.63%), while Battle Royale had the lowest (45.61%). HRV score stayed close (87–88) across all genres.

### 8.5 Blue-Light Filter Use Is Common

About 65% of players use a blue-light filter, showing it's a widely adopted habit among this player base.

### 8.6 Business / Practical Implications

Taken together, these findings support building sleep and recovery guidance into training programs, treating reaction time as a more reliable performance signal than raw APM, and monitoring energy drink intake for its physiological load rather than any performance upside. They also suggest genre-specific coaching should focus more on win-rate patterns than on recovery metrics, which stay fairly stable across genres.

# 9. Power BI Dashboard

To bring everything together, we built an interactive Power BI dashboard that presents the major findings from the analysis in a clear, visual way. Rather than just reading about the results, the dashboard lets you explore them directly — by genre, sleep group, APM group, and match outcome.

Dashboard Title: Game Sense – Performance & Health Analytics

## 9.1 Dashboard Features

- Total Players, Win Rate %, Loss Rate % – headline KPI cards
- Average APM, Average Reaction Time, Average Sleep – headline performance metrics
- Win Rate by Game Genre, Sleep Group, and APM Group – comparative bar charts
- APM vs. Reaction Time – scatter plot colored by match outcome
- Win Rate by Energy Drinks and by Sleep Group – trend line charts
- Average HRV Score and Heart Rate by Sleep Group, Energy Drinks, and Playtime Group
- Blue Light Filter Usage – usage-share donut chart
- Players by Age Group – demographic breakdown
- Shared filters for Game Genre, Blue Light Filter, Match Outcomes, and Sleep Groups

## 9.2 Dashboard Purpose

The dashboard takes the analytical results and turns them into an interactive, coach-and-player-friendly view. It makes it easy to quickly see overall performance, spot which health and behavioral factors line up with winning, compare genres, and explore the relationship between recovery and performance — all in one place.

## 9.3 Dashboard Screenshots




*Figure 1: Player Performance Overview*




*Figure 2: Performance Analysis*




*Figure 3: Health & Gaming Behaviour Analysis*

## 10. Conclusion

In short, this project cleaned and analyzed 250,000 esports match sessions using Python, Pandas, and Power BI to uncover what's really linked to winning. Sleep and reaction time turned out to be more consistently tied to win rate than raw APM, energy drink use drives up heart rate without a clear performance payoff, and genre shifts win rate modestly while leaving recovery metrics largely unchanged. The resulting Power BI dashboard brings all of these findings together into one easy-to-explore view that can support better training, recovery, and coaching decisions.

# 11. Future Improvements and Limitations

## 11.1 Limitations

- **Descriptive Analysis:** this project focuses on descriptive and exploratory analysis and doesn't include any predictive modeling.
- **Correlation vs. Causation:** relationships identified (e.g., sleep and win rate) are correlational, not causal — other unmeasured factors could be involved.
- **No Longitudinal Tracking:** the dataset captures single-session snapshots rather than tracking the same players over time, so it can't show how habits change individual performance longitudinally.
- **Genre Scope:** the dataset covers a fixed set of game genres, so findings may not generalize to other genres or esports titles.

## 11.2 Future Improvements

- Add player-level longitudinal tracking to see how changes in habits affect the same player's performance over time
- Introduce predictive modeling for win probability based on health and performance inputs
- Add real-time biometric tracking during live matches
- Bring in additional datasets such as team, tournament, or coaching data
- Automate data refresh and dashboard updates
- Enhance the Power BI dashboard with more coach-focused views and drill-down capabilities

## 12. Project Developed By

| Details | Information |
|---|---|
| Project Title | Game Sense – Esports Performance & Gaming Health Analytics |
| Project Type | Data Analytics Project |
| Team Name | Data Dynamos |
| Team Members | Teja Sri, Ajay Kumar, Lahari |
| Course / Program | Data Analytics |
| Institution | TEKS Academy |
| Tools Used | Python, Pandas, Matplotlib, Seaborn, Power BI |
| Date | September 7, 2026 |

## 13. References and Dataset Source

### 13.1 Dataset Source

We sourced the Esports Gaming Biometrics Dataset used for this project from Kaggle.

### 13.2 Project Resources

- GameSense.ipynb – the Python/Pandas notebook used for data cleaning, transformation, validation, and exploratory analysis
- Power BI report – used to build the interactive performance and health dashboard

### 13.3 Technologies Referenced

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Microsoft Power BI

**END OF PROJECT DOCUMENTATION**